Thesis title

A thesis submitted in partial fulfilment of the requirements for the PhD degree.

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June 2025

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A Third University

Abstract

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The work, text, and figures presented in this thesis are my own unless otherwise stated. Where work from publications on which I am co-author is adapted, this is made clear in the heading of the corresponding chapter.

Acknowledgments

Example quote

Book title, A. Uthor

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List of publications

Major contributions

- **J. Dranczewski**, A. Fischer, P. Tiwari, M. Scherrer, D. Saxena, H. Schmid, R. Sapienza, K. Moselund, "Plasma etching for fabrication of complex nanophotonic lasers from bonded InP semiconductor layers" *Micro and Nano Engineering*, vol. 19, p. 100196, Jun. 2023.

- D. Saxena,* A. Fischer,* **J. Dranczewski**, W. K. Ng, N. V. Trivino, H. Schmid, T. V. Raziman, A. Arnaudon, M. Barahona, O. Hess, K. Moselund, R. Sapienza, "Designed Semiconductor Network Random Lasers," *Laser & Photonics Reviews*, vol. 19, no. 1, p. 2400623, 2024.

- **J. Dranczewski**,* W. K. Ng,* A. Fischer,* T. V. Raziman, D. Saxena, T. Farchy, K. Stenning, J. Peters, H. Schmid, W. R. Branford, M. Barahona, K. Moselund, R. Sapienza, J. C. Gartside, "Retinomorphic Machine Vision in a Network Laser," *arXiv*, Aug. 2025. *[manuscript in review]*

Minor contributions

- A. Fischer, T. V. Raziman, W. K. Ng, J. Clarysse, D. Saxena, **J. Dranczewski**, S. Vezzoli, H. Schmid, K. Moselund, R. Sapienza, "Controlling lasing around exceptional points in coupled nanolasers," *npj Nanophotonics*, vol. 1, no. 1, pp. 1–6, May 2024.

- A. Fischer,* T. Severs Millard,* X. Xiao, T. V. Raziman, **J. Dranczewski**, R. C. Schofield, H. Schmid, K. Moselund, R. Sapienza, R. F. Oulton, "Surface Lattice Resonance Lasers with Epitaxial InP Gain Medium," *ACS Photonics*, vol. 11, no. 10, pp. 4316–4322, Oct. 2024.

*Authors contributed equally.

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Acronyms

	Computing
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
GPU	graphics processing unit
GUI	Graphical User Interface
LLM	Large Language Model
ML	Machine Learning
MLP	multilayer perceptron
RAM	Random Access Memory
RGB	red, green, and blue
	Fabrication
ALD	Atomic Layer Deposition
BOX	burried oxide
DoE	Design of Experiment
DWB	Direct Wafer Bonding
FIB	Focused Ion Beam
HSQ	hydrogen silsesquioxane
ICP	Inductively Coupled Plasma

MOCVD	metalorganic chemical vapour deposition
PECVD	plasma-enhanced chemical vapour deposition
QD	quantum dot
QW	quantum well
RF	Radio Frequency
RIE	Reactive Ion Etching
TASE	Template-Assisted Selective Epitaxy
VCSEL	Vertical Cavity Surface Emitting Laser

Chapter 1
Example chapter

For the work presented in this chapter I have contributed <X>. Most figures in this section, as well as parts of the relevant discussion, are adapted from the following publication, plotting for which has been done by me, and a couple of other people:

J. Dranczewski,* W. K. Ng,* A. Fischer,* T. V. Raziman, D. Saxena, T. Farchy, K. Stenning, J. Peters, H. Schmid, W. R. Branford, M. Barahona, K. Moselund, R. Sapienza, J. C. Gartside, "Retinomorphic Machine Vision in a Network Laser," *arXiv*, Aug. 2024. *[manuscript in review]*

*Authors contributed equally.

Abstract

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1.1 Example section

- You can write notes to yourself in "outline" environments. They act like LaTeX's "itemize", but are coloured blue, and you can easily hide them from the final document once you're done with a specific outline by changing the environment name to "outlineDone".
- I also provide a number of potentially useful icons that can be used to annotate these outlines:
  - Checkmarks, for points you've covered ✓

  - Hourglass for things that are in progress
  - Paper, for things to be adapted from papers
  - Globe, for things that need to be referenced or researched from external sources
  - Notion, as that's where I was tracking figure progress

- You can use these anywhere in the text just by using the commands listed in the source above: ✓

Example citation[1] – citations appear in the upper right by default, if you need to reference a citation in the main text you can use Ref. [1]. Note the ~ symbol – it forms a non-breaking space between "Ref." and the citation number, meaning that they will always appear on the same line. You can similarly reference chapters and sections, for example Chapter 1.

Here's an example figure: Fig. 1.1

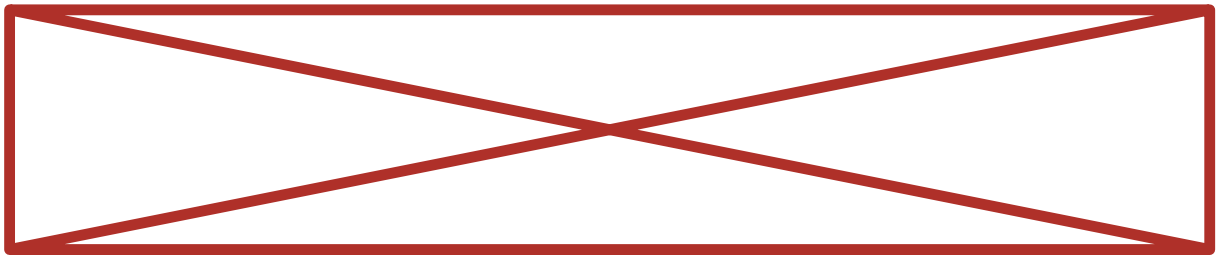

FIGURE 1.1: Example figure description. **a** This describes part *a* of the figure. **b** This describes part *b* of the figure.

Here's an example of a narrow figure with the caption to the side: Fig. 1.2

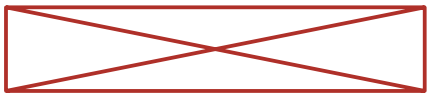

FIGURE 1.2: Narrow figure caption.

Examples of acronym usage: first instance – *Central Processing Unit* (CPU), second instance – CPU. Forced full: *Central Processing Unit* (CPU).

Bibliography

[1] J. Dranczewski, A. Fischer, P. Tiwari *et al.*, "Plasma etching for fabrication of complex nanophotonic lasers from bonded InP semiconductor layers," *Micro and Nano Engineering*, vol. 19, p. 100196, Jun. 2023.